# KAVIN PASUPATHY

Software Engineer · Full-Stack (MERN / Next.js / TypeScript) · AI Agents & Automation
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## PROFESSIONAL SUMMARY

Software Engineer with ~2 years of experience building production web applications with the **MERN stack, Next.js, TypeScript, and PostgreSQL**, and delivering **AI-agent and workflow-automation solutions** for client projects. Own features end-to-end — API and database design through Dockerized deployment — with hands-on experience integrating **LLM APIs (Claude)**, tool/function calling, RAG with pgvector, and agent observability tooling (**Langfuse**). Ranked 1st in undergraduate Computer Science department.

## TECHNICAL SKILLS

**Frontend:** React.js, Next.js, TypeScript, JavaScript (ES6+), HTML5, CSS3, Responsive UI
**Backend:** Node.js, Express.js, REST APIs, JWT Authentication, Nodemailer, Multer
**Databases & ORM:** PostgreSQL, Supabase, pgvector, MongoDB (Mongoose), Drizzle ORM, Prisma ORM, SQL
**AI & Automation:** AI Agents, Agentic Workflows, AI Agent Orchestration, LLM Integration (Claude API, Anthropic SDK), Tool/Function Calling, Model Context Protocol (MCP), RAG, Vector Search, Embeddings (Voyage, Gemini, OpenAI), Agent Skills, Prompt Engineering, Prompt Caching, Vercel AI SDK, Langfuse (LLM Observability), OpenTelemetry
**Tools & DevOps:** Git, GitHub, GitHub Actions (CI/CD), Docker, Docker Compose, AWS S3, Vercel, Zod, VS Code, Claude, Cursor
**Programming:** TypeScript, JavaScript, Python, SQL

## EXPERIENCE

**Junior Software Engineer — Belsterns Technologies** Chennai | Jan 2025 – Present

- Develop and ship full-stack features for client web applications using **Next.js, TypeScript, and PostgreSQL**, deployed in Dockerized environments.
- Develop features for **Eztest**, an online testing management platform built with Next.js and PostgreSQL; build AI-agent based task orchestration and intelligent process automation for business workflows.
- Design and build **AI-agent workflows** that automate client reporting and business processes — integrating **Claude LLM APIs** with tool/function calling, orchestrating multi-step agent tasks, and monitoring agent behavior with **Langfuse**.
- Lead client modules end-to-end: requirements analysis, API and database design, implementation, and release, collaborating with cross-functional teams to meet delivery timelines.
- Reduced manual reporting effort for clients by replacing repetitive workflows with automated agent pipelines.

**Software Development Engineer Intern — Belsterns Technologies** Chennai | Sep 2024 – Jan 2025

- Built and optimized responsive web applications in the **MERN stack and Next.js** for client-facing projects.
- Contributed to client-based projects using Next.js; **promoted to full-time Junior Software Engineer within 4 months**.

## PROJECTS

**BlackSpace AI — Multi-Agent LLM Platform** Next.js · TypeScript · Claude API · Vercel AI SDK · pgvector | Aug 2026 | GitHub

- Built a general-purpose **AI agent platform** where Claude runs an agentic **tool-calling loop** (call tools → read results → repeat) with streamed reasoning, tool status and source citations; agents are config profiles (prompt, tools, step limit, reasoning effort), with 5 built in: general, knowledge base, researcher, data analyst and offline.
- Implemented **RAG** on Postgres/Supabase with **pgvector**: paragraph-aware chunking with overlap, cosine-similarity search with a calibrated relevance floor, and pluggable **embeddings** (Voyage, Gemini, OpenAI) that switch providers without a migration.
- Reduced LLM cost on long agent runs with **prompt caching** and server-side **context compaction**; added per-turn cost tracking and per-user monthly spend limits, with full run tracing in **Langfuse** via OpenTelemetry.
- Secured the app with scrypt password hashing, server-side hashed sessions, role-based admin access and repository-level data ownership; **Drizzle ORM** migrations, S3 presigned uploads and Dockerized local Postgres.

**Home Rental Platform — MERN Stack** Jul 2024

- Built a rental booking system with full CRUD, availability calendar, **JWT authentication**, automated email notifications (Nodemailer), and image uploads (Multer).

## EDUCATION

**Master of Computer Applications (MCA) — Kongu Engineering College, Perundurai** 2023 – 2025 | CGPA: 8.4

**B.Sc. Computer Science — Government Arts & Science College** 2020 – 2023 | CGPA: 7.57

- **Achievement:** Ranked 1st in the Undergraduate Computer Science Department.

## CERTIFICATIONS

- Claude Certified Architect – Foundations — Anthropic | Issued Sep 17, 2026